DPP — DEPARTMENTAL POLICY AND PROCEDURE

TITLE:	MANAGEMENT OF NEONATAL HYPOGYLCEMIA
DEPARTMENT:	NEONATAL INTENSIVE CARE UNIT
APPLIES TO:	NICU PHYSICIANS AND NURSES

CODE AND VERSION NUMBER	
DPP-NICU-0020 (v3)	
LAST VERSION APPROVAL DATE:	03 / 07 / 2023
NUMBER OF PAGES:	4

	NAME	POSITION	SIGNATURE
PREPARED BY:	MS. KRISTINE P. MAGNETICO	UNIT QUALITY REPRESENTATIVE	
	MS. SHATHA MOHAMMED	NICU HEAD NURSE	
REVIEWED BY:	MEDICAL EXECUTIVE COMMITTEE		
	DR. ABDULKARIM IBRAHIM	HEAD OF NICU	
	MS. MERRY JOY H. GALLEBO	NURSING QUALITY COORDINATOR	
	MR. KHALID GHURMULLAH ALGHAMDI	QPS DIRECTOR	
APPROVED BY:	MS. HAIFA SAAD MISFER	DIRECTOR OF NURSING	
	DR. RAKAN KHALIL ALGHAMDI	AED FOR MEDICAL SERVICES	

APPROVED DATE	EFFECTIVE DATE	TYPE OF REVISION	REVIEW DATE
09 / 02 / 2026	23 / 02 / 2026	TRIENNIAL	08 / 02 / 2029

CONTENTS

1. PURPOSE	4. POLICY	7. REFERENCE/CROSS-REFERENCES
2. DEFINITION & ABBREVIATION	5. PROCEDURE	8. ATTACHMENT
3. RESPONSIBILITY	6. MEASUREMENT	9. REVISION CLAUSE

1. PURPOSE

1.1. To provide guidelines on how to assess and manage newborn with neonatal hypoglycemia to prevent brain damage.

2. DEFINITION & ABBREVIATION

2.1. **Hypoglycemia** – a serum glucose value of <45 mg/dl in a term or preterm neonate which suggest clinically significant hypoglycemia are non-specific and include jitteriness, irritability, high pitched cry, cyanosis, apnea, seizures, lethargy, hypotonia or poor feeding.

2.2. **Neonatal Hypoglycemia** – newborn's blood glucose level is too low to support normal brain function.

2.3. **BSL** – Blood Sugar Level, measured by a blood glucose monitor (glucometer) and reagent strip in birth center and postnatal wards.

2.4. **TBG** – True Blood Glucose, as measured on a blood gas analyzer in NICU or by laboratory measurement.

2.5. **EMR** – Electronic Medical Records

2.6. **MCC** – Medica Cloud Care system / **HIS** (Hospital Information System)

2.7. **NICU** – Neonatal Intensive Care Unit

3. RESPONSIBILITY

3.1. **Head of NICU**

3.1.1. Ensures the strict adherence of all staff to this policy and procedures.

3.2. **Physicians (Consultant, Specialists and Residents)**

3.2.1. Provides the orders in the physician's order sheet at EMR.

3.3. **Nurses (Head Nurse, Charge Nurse and Staff Nurse)**

3.3.1. Verifies and implements physician's order.

3.3.2. Assess, reassess and monitor patient with hypoglycemia.

4. POLICY

4.1. Physicians will follow the recommended treatment and considerations in the management of hypoglycemia as stated below.

4.1.1. Monitor infants with risk factors for hypoglycemia including maternal indications

4.1.1.1. Mother who has diabetes

4.1.2. Infant indications

4.1.2.1. Small for gestational age infants (10th centile)

4.1.2.2. Low birth weight infants (<2.5 grams)

4.1.2.3. Clinically wasted regardless of birth weight

4.1.2.4. Macrosomic or large for gestational infants (>90th centile)

5. PROCEDURE

5.1. The physician and nurse on duty will assess the patient with potential risk and predisposing factors to develop hypoglycemia which follows:

5.1.1. Impending birth of an infant of a mother with diabetes.

5.2. Management of infants with risk factors for or diagnosis of hypoglycemia in OBSN:

5.2.1. Measure BSL or TBG on blood gas analyzer.

5.2.1.1. Within one hour of arrival in nursery (or as ordered by pediatric specialist).

5.2.1.2. If TBG ≥ 2.6 mmol/L, repeat before next feed, or 3 hourly if infant nil orally.

5.2.1.3. If TBG < 2.6 mmol/L, repeat TBG in 1 hour

5.2.1.4. Measure TBG before feeds if feed due

5.2.1.5. Glucose monitoring may be discontinued when 3 consecutive TBG readings ≥ 2.6 mmol/L.

5.3. Physician will enter orders in the doctors order sheet in MCC.

5.4. Nurse on duty will verify physician's order and implement treatment as prescribed.

5.5. Re-assessment of patients receiving treatment for hypoglycemia shall be performed.

5.6. Document all the procedures done in the patient's electronic medical records (EMR).

6. MEASUREMENT

6.1. 100% of the staff is knowledgeable and adhere to this policy and procedures.

7. REFERENCES /CROSS- REFERENCES

7.1. PMSH-DPP-NICU-0023(v3): Management of Neonatal Hypoglycemia, 03/07/2023, p. 1-3

7.2. CBAHI Standards 3rd edition, 01/01/2016, NICU.10, NICU.12 p.126-127

7.3. Harriet Lane 17th edition, 2009

7.4. NGHA-Parenteral Therapy Manual, 2006

7.5. HSC Handbook of Pediatrics 10th edition, 2004

7.6. Pediatric Endocrinology by Sperling 2nd edition, 2002

7.7. Pocket pediatrics by O'Callaghan and Stephenson, 2001

8. ATTACHMENT

8.1. None

9. REVISION CLAMSE

9.1. Changes to the hospital policy and procedure template based from the Al-Baha Health Cluster logo, font face and design.

9.2. Change of code from DPP-NICU-0023 to DPP-NICU-0020